

SUMMARY REPORT

APPLE STOCK ANALYSIS USING PYTHON

Dataset Overview

This project analyzed historical Apple Inc. (AAPL) stock market data from June 2021 to June 2026 using advanced Python techniques for data transformation, time-series analysis, and feature engineering. The dataset contained 1,255 trading records and included the following variables: Date, Open Price, High Price, Low Price, Close Price, and Trading Volume.

The analysis was conducted using Python libraries such as Pandas, NumPy, Matplotlib, and Seaborn to clean, transform, analyze, and visualize stock market trends.

Data Transformation Techniques Applied

Several preprocessing and transformation techniques were applied to prepare the data for analysis. The Date column was converted into a datetime format to enable time-series operations, while stock prices and trading volume were converted into appropriate numeric data types. The dataset was cleaned by removing unnecessary rows, checking for missing values, and organizing records chronologically.

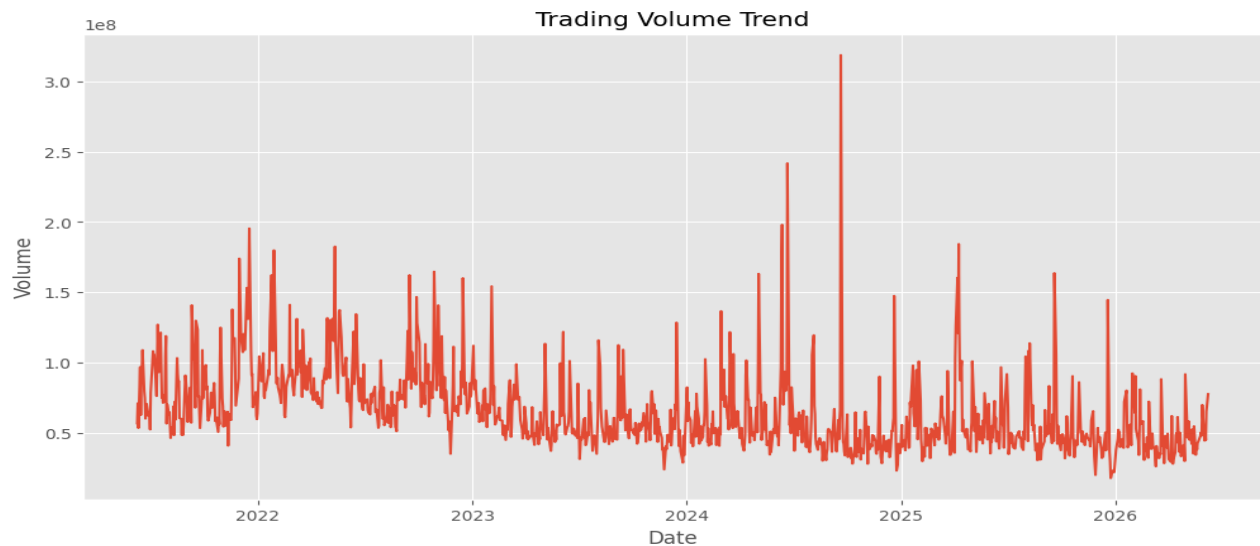
Additional transformations included extracting Year and Month information from the Date field, grouping data by month, and calculating average closing prices and trading volumes. These steps improved data quality and enabled more meaningful analysis of stock performance over time.

TRENDS IDENTIFIED

1. **Stock Closing Price:** The Stock Closing Price Trend revealed strong long-term growth in Apple's stock performance. The closing price increased from approximately \$124 in 2021 to over \$315 in 2026, demonstrating significant appreciation in market value despite periodic fluctuations.

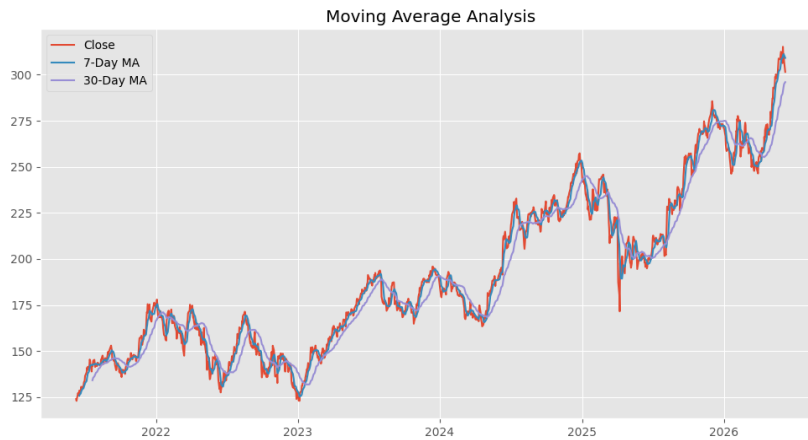


2. **Trading Volume:** The Trading Volume Trend showed substantial variation in investor activity throughout the period. Average daily trading volume was approximately 65.35 million shares, while peak trading volume exceeded 318 million shares during periods of heightened market activity.

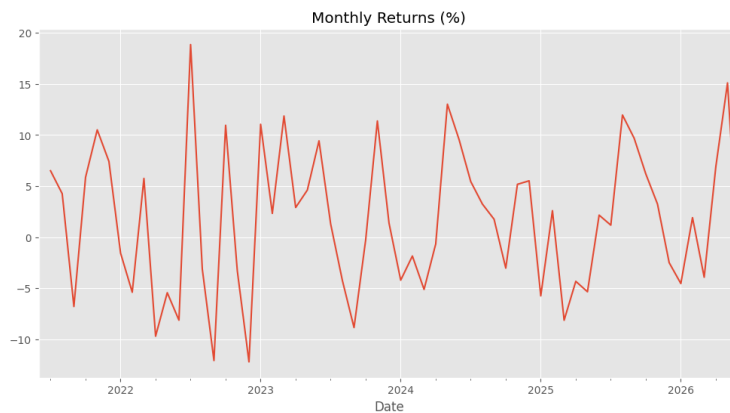


3. **Moving Average Analysis:** The Moving Average Analysis compared the 7-day and 30-day moving averages with the closing price. The 7-day moving average reflected short-term market movements, while the 30-day moving

average highlighted longer-term trends. Both indicators confirmed the overall upward trajectory of Apple's stock.



4. Monthly Return Analysis: Monthly Returns Analysis showed fluctuations in stock performance, with monthly returns ranging from approximately -12% to +19%. This indicates that although the stock delivered strong long-term growth, investors experienced periods of both gains and losses.



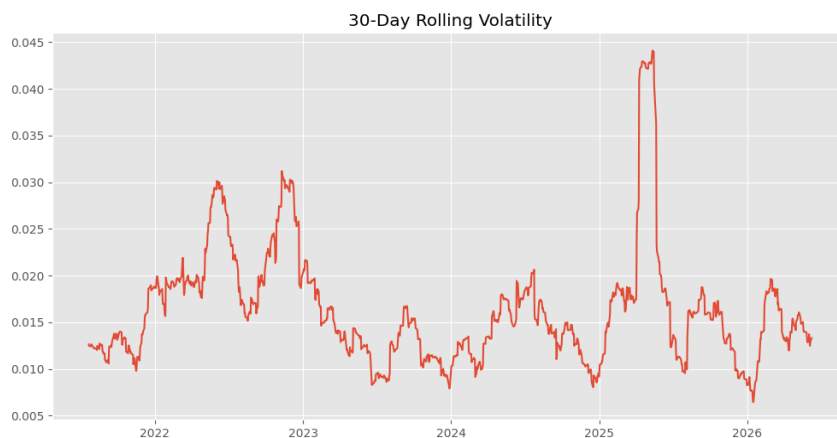
FEATURES CREATED

To enhance the analysis, several new features were engineered from the original dataset:

- Daily Price Change (Closing Price – Opening Price)

- Percentage Price Change
- 7-Day Moving Average
- 30-Day Moving Average
- Daily Return
- Monthly Returns
- 30-Day Rolling Volatility

The volatility measure was particularly useful in identifying periods of increased market uncertainty and risk.



KEY INSIGHTS

The analysis produced several important insights. Apple demonstrated strong long-term growth, with stock prices increasing substantially over the five-year period. The average closing price during the analysis period was approximately \$191.72, while the highest recorded closing price exceeded \$315.

Trading activity remained consistently strong, with average daily trading volume above 65 million shares. Monthly returns highlighted both profitable and challenging periods, reflecting the natural volatility of financial markets. The moving average indicators effectively identified overall market direction while reducing the impact of short-term price fluctuations. Additionally, volatility

analysis revealed periods of increased risk, emphasizing the importance of monitoring market uncertainty when making investment decisions.

RECOMMENDATIONS

Based on the findings, investors should monitor moving average indicators to identify potential trend changes and investment opportunities. Periods of increased volatility should be approached cautiously, as they may indicate higher market risk. Trading volume should be analyzed alongside price movements because unusually high volume often signals stronger market sentiment and significant price changes.

For future analysis, additional technical indicators such as Relative Strength Index (RSI), Bollinger Bands, and Exponential Moving Averages (EMA) could be incorporated to provide deeper insights and improve forecasting accuracy.

CONCLUSION

This project successfully applied advanced Python techniques to transform, analyze, and visualize historical Apple stock market data. Through data cleaning, feature engineering, and time-series analysis, valuable insights were generated regarding stock price trends, trading activity, returns, and volatility. The results demonstrate how Python can be used to convert raw financial data into meaningful information that supports data-driven investment analysis and decision-making.

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